

Electronics & Circuits Bench — Cheat Sheet

Last-minute recall reference. For the reasoning behind each row, see [LESSON.pdf](#) or the interactive tour.

CONCEPT	FORMULA / RULE
Ohm's law	$V = I \times R$
Power	$P = V \times I$ — check the resistor's wattage rating, not just its ohms
Series resistance	$R_{\text{total}} = R1 + R2 + \dots$
Voltage divider output	$V_{\text{out}} = V_{\text{in}} \times R2 / (R1 + R2)$ — sags under load
LED with no series resistor	Runaway current, fails almost instantly — LEDs aren't resistors
LED series resistor	$R = (V_{\text{supply}} - V_{\text{forward}}) / I_{\text{desired}}$
Measuring voltage	Meter in parallel , circuit powered
Measuring current	Meter in series — break the circuit and insert it
Continuity	Power off , beeps below a few ohms
Blown meter fuse	Usually leads left in the 10A jack while measuring voltage
Good solder joint	Concave, shiny fillet — it wetted the pad
Cold joint	Dull, convex ball — add flux, not more heat or time
Water-soluble flux	Must be washed off, or it corrodes the joint
Rosin-core solder	Flux built into the core — usually enough on fresh boards
Tinning the tip	Coat the iron tip in fresh solder so it actually transfers heat
Fault-finding order	Power at the component → ground → orientation → joints
Debugging rule	One change, one measurement — every time
Intermittent fault (press-to-work)	Almost always a cold/cracked solder joint